

Computer and Network Security

Tutorial Worksheet B-02

Description. This tutorial worksheet covers modern cryptography: :

- RSA cipher
- El Gamal cipher
- Diffie-Hellman key establishment
- Elliptic Curve Cryptography

RSA Cipher

Exercise 1

Zoubir lent his debit EMV card to Massoud. Massoud needed the card to perform an urgent transaction. To use the card, Massoud needed both the card and its 3-digit secret code. For security reasons, Zoubir decides to send the code to Massoud using RSA encryption. The public key of Zoubir is $(3, 55)$ and the one of Massoud is $(11, 1343)$.

1. Compute the private key of Zoubir and Massoud, respectively.
2. As mentioned above, Zoubir encrypts the code using Massoud's public key. What would be the ciphertext of the plaintext code 389?
3. To prove its identity and enforce non-repudiation, Zoubir decides to sign the encrypted code before sending it to Massoud. Massoud receives the value 18. Check that.

Exercise 2

A website www.gotpumped.com earns the public key $(e_w, n_w) = (5, 65)$. Its private key d_w is secretly stored on the server. The administrator of the website uses a smartcard to get authenticated to the website server. The administrator smartcard public key is $(e_a, n_a) = (7, 77)$. Its private key d_a is also secretly stored on the card (tamper-resistant).

1. Compute the private key of both parties.
2. The website has to transmit the value 3 to the smartcard. What would be the encrypted value for the message 3. How does the smartcard retrieve the value of 3?
3. To ensure both authenticity and confidentiality, the website sends the tuple (c, s) to the smartcard, where c is the ciphertext of the message $m = 3$, and s is the signature of the message m encrypted using the administrator's public key. What is the value of the tuple? How does the smartcard retrieve the message $m = 3$?

4. If the website performs the previous cryptographic operations in the inverse order, would it be correct? Justify.

Exercise 3

Consider the scenario of RSA digital signature:

1. Compute the modulus n and the Euler's totient $\Phi(n)$ when the prime numbers $p = 17$ and $q = 23$.
2. Among the following two public exponents, $e = 11$ and $e = 13$, which one allows to use RSA?
3. Compute the private key d for the suitable value of e .
4. If Zoubir wants to sign the message $m = 100$ before sending it to Abbas, what would be the value of signature?
5. How does Abass verify the signature?

Exercise 4

Consider an RSA cryptosystem with $n = 35$. Alphabetical letter are encoded following the below table:

A	B	C	D	E	F	G	H	I	J	K	L	M
1	2	3	4	6	8	9	11	12	13	16	17	18
N	O	P	Q	R	S	T	U	V	W	X	Y	Z
19	22	23	24	26	27	29	31	32	33	33	33	33

1. Compute $\phi(n)$. The value of e was set to 5.
2. Compute the value of d .
3. Decrypt the phrase "IA MEUSEBEUAQXE MALE" that was encrypted using the public key $(5, 35)$.

Exercise 5

During a secure communication, Zoubir sent a message (secret code) to Massoud. The message contains characters encoded in 16 bits (i.e., each character is encoded using 4 hexadecimal digits). Abbas eavesdropped the communication between Zoubir and Massoud, and managed to capture the following ciphertext 0x23C8 0x004D 0x0591 0x0002 0x0000. Knowing that the encryption was performed using RSA with parameters $p = 101$ and $q = 113$, and that the public key of Massoud was $e_M = 7467$, what was the secret message if the alphabetical letters were encoded as A=0, B=1, ..., Z=25?

Exercise 6

We want to use RSA cipher with modulus parameter $n = 365$. How many valid public keys e can be used?

Exercise 7

Consider the RSA cipher where Zoubir and Massoud decided to generate their public and private key using the same modulus, such that $\gcd(e_Z, e_M) = 1$. Abbas, the MITM, managed to capture two ciphertexts, $C_Z = (m)^{e_Z}[n]$ and $C_M = (m)^{e_M}[n]$. Moreover, Abbas computes $x_1 = e_Z^{-1}[e_M]$ and $x_2 = (x_1 \cdot e_Z - 1)/e_M$. Illustrate how Abbas can compute m using C_Z, C_M, x_1 , and x_2 . Compute m if $n = 18721$, $e_Z = 43$, $e_M = 7717$, $C_Z = 12677$, and $C_M = 14702$. What do you conclude?

Diffie-Hellman Key Establishment

Exercise 8

Zoubir and Massoud use DH protocol with modulus $n = 11$ and cyclic-group generator $g = 2$. Knowing that both Zoubir and Massoud secretly and randomly generate their private keys as $d_Z = 7$ and $d_M = 8$, respectively, compute the shared secret key k that both Zoubir and Abbas compute at the end of the DH protocol.

Exercise 9

Zoubir and Massoud want to establish a shared secret key k . To that end, they decide to use the DH protocol. Both Zoubir and Massoud secretly and randomly generate their private keys as d_Z and d_M , respectively. During the execution of the protocol, Abbas, a man-in-the-middle, captures $\alpha = g^{d_Z}[n]$ and $\beta = g^{d_M}[n]$. The public parameters of the protocol are known to be $n = 71$ and $g = 7$.

1. Give two possible tuples (α, β) such that the shared secret key $k = 1$.
2. From some sources, Abbas got to know that $\alpha \times \beta = 7[71]$. Provide two possible tuple of (d_Z, d_M) that make Abbas happy.

El Gamal

Exercise 10

Consider El Gamal cipher with the following parameters: $n = 59$, $g = 2$, and $P = 56$.

1. Verify that $s = 21$ is the private key for the El Gamal public key (n, g, P) .
2. Computer the ciphertext for $m = 7$ using the above public key. Why is the ciphertext not unique?
3. Illustrate the steps to decrypt the ciphertext.

Exercise 11

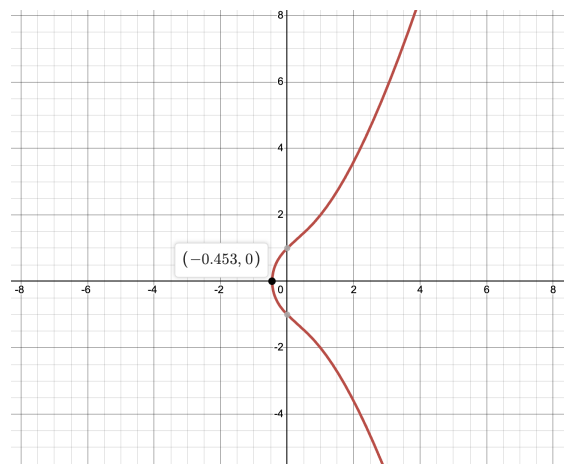
Consider El Gamal cipher with parameters (n, g, P) .

1. Let $c = (a, b)$ be a ciphertext of a message m . Assuming that Abbas can obtain the plaintext of a ciphertext $c' \neq c$, show that he can find the original message m .
2. Let $c_1 = (a_1, b_1)$ and $c_2 = (a_2, b_2)$ two ciphertexts of the two distinct messages m_1 and m_2 , respectively. Show that Abbas can encrypt another message m_3 .

Elliptic Curve Cryptography

Exercise 12

Consider the elliptic curve E' defined by the following plot:



1. Find the parameters of the curve and give the Weierstrass equation of the curve.
2. Can this curve be used in cryptography?
3. Define the curve E over $\mathbb{Z}_7^2$.
4. Verify that the point $P = (1, 5)$ lies on E .
5. Find all points of E . Compare the number of points with the Hasse's theorem.
6. For each point $P \in E$, compute $-P$ and check that it lies on the curve as well.
7. Compute the sum $P + Q = (0, 6) + (0, 1)$.
8. Compute the sum $P + Q = (0, 6) + (1, 2)$.
9. Compute the point $2P, 3P, 4P, 5P, 6P, 7P$ where $P = (0, 1)$. What do you observe?

Exercise 13

Abass is trying to disclose an RSA private key component d (the decryption exponent) using side-channel attack patterns. He managed to obtain partial information about d from different computations that d underwent in a computer system. The power analysis of the CPU revealed that d was congruent 158 modulo 173, the timing analysis of the CPU revealed that d was congruent 471 modulo 577, and from memory leakage, it was revealed that d was congruent 383 modulo 811. The exponent d is related to the public exponent $e = 38320791$ and the modulus $n = 207310381$. Assuming Abass does not have the resources to factorize n into two prime numbers, help them disclose the key d .

Exercise 14

Answer the following questions:

1. In ElGamal, why can the destination decrypt an encrypted message sent to them?
(1) They know the private key, (2) They can figure out the value of k , (3) The value of k is not necessary, and (4) the value of k is implicit.
2. In the Affine cipher, how many valid values of the parameter ' a ' exist, if the system operates modulo 255? 128, 155, 32640, or 32512.
3. Consider the following set T of RSA public keys (n, e) . Which among the keys are valid for RSA application? $T = \{(60, 11), (391, 11), (391, 13), (77, 7)\}$
4. In ElGamal, if a man-in-the-middle learns the value of k that was used to generate the ciphertext (c_1, c_2) of a plaintext m , then: (1) They can do none, (2) They can decrypt the message, and (3) they can learn the secret key.
5. The ElGamal public key of Zoubir is $(p, g, h) = (101, 2, 11)$. Find the plaintext m of the ciphertext $(c_1, c_2) = (64, 79)$ that was sent to Zoubir by Massoud without cracking the private key of Zoubir. Disclose the key of Zoubir.
6. Suppose that you are using an exponential version of ElGamal, where instead of encrypting the message m , you encrypt the message g^m , where $m \in \mathbb{Z}_p$. This scheme is used in an application where the value 10 encodes the keyword "ACCEPT", the value 50 encodes the keyword "DECLINE", and the other values are "ERROR". Demonstrate that if you intercept a ciphertext $c = (c_1, c_2)$ of an unknown message m , you can still tamper-with the original message m and make it another m' without the need to decrypt the original ciphertext to m .
7. Zoubir and Massoud used Diffie-Hellman to derive a secret session key k . They used $g = 5$ and $p = 47$. The exchanged values were $X = 14$, $Y = 8$? What is the key?

x	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
g^x	5	25	31	14	23	21	11	8	40	12	13	18	43	27	41	17	38	2	10	3

Exercise 15

Consider the following protocol where Zoubir and Massoud use their respective asymmetric keys:

1. Zoubir sends to Massoud $K^a[p]$,
 2. Massoud replies with $(K^a)^b[p]$, and
 3. Zoubir sends back $((K^a)^b)^{a^{-1}}$.
- What information is shared among Zoubir and Massoud, and which one is not?
 - Describe how the protocol works and explain how Massoud finds K .
 - Run a MITM attack on the protocol, assuming that asymmetric keys were not properly used by Zoubir and Massoud.

Exercise 16

Abass knew that Zoubir and Massoud used DH to derive a secret k . He was able to capture the key establishment phase of the protocol. Abass has a friend of his, Hamid, who works in Japan and has access to the Fugaku, a 442 PFLOPS supercomputer. We consider that, using the best modular exponentiation algorithms, the supercomputer may hit 10^{13} exponentiations per second.

1. Abass assumed that Zoubir and Massoud used a 2048-bit order multiplicative cyclic group with generator g . He wanted to use the Fugaku to find the private keys of Zoubir and Massoud, i.e., brute-forcing the discrete logarithm problem. In the worst case scenarios, how long will it take for Abass to find the key of either parties?
2. Hamid got fired, and access to the Fugaku was lost. Abass decided to use an alternative approach where he learned that Zoubir and Massoud used a multiplicative group of order $p - 1 = 2310$. He also heard that the generator g of the group was either 4, 7, 11, or 13. Furthermore, after talking to some cryptographers, it was suggested to Abass to apply Pohlig-Hellman attack (a.k.a. PH attack) to passively recover the private exponent of Zoubir, which he intercepted as $g^x = 2227 [p]$.